Before the Command Was Spoken: Law as the Extended Phenotype of Pre-Linguistic Norms

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Abstract

This paper argues that law is not a human invention ex nihilo, but the *extended phenotype* of pre-linguistic normative patterns that evolved to stabilize cooperation in early human groups. Drawing on Dennett's principle of "competence without comprehension," Dawkins' theory of the extended phenotype, and Richerson & Boyd's models of gene-culture coevolution, I propose that heteronomous norms predated prescriptive language and exerted selective pressure for its emergence. Using cross-linguistic evidence from deontic lexicons, archaeological markers of prehistoric social organization, primatological studies of third-party enforcement and coalition formation, and ethnographic documentation of ritual transmission, I demonstrate that proto-norms—initially encoded in embodied practice, partner selection, and collective sanctioning—created selection pressures for linguistic structures capable of abstracting, generalizing, and transmitting obligations across contexts. Experimental studies across great apes and New World monkeys (*Pan troglodytes* N=46, *Cebus apella* N=6) show systematic absence of normative enforcement: chimpanzees exhibit 0% third-party punishment (rejection rates 5-14% vs. 50% in humans; Riedl et al., 2012), while capuchins punish inequity (25.7%) but remain insensitive to intentionality (Leimgruber et al., 2016), demonstrating outcome-based rather than norm-based responses. Archaeological analysis reveals conspicuous absence of exclusionary burial practices in Upper Paleolithic contexts (0.67%, 1/150 burials) versus systematic emergence in Neolithic/Chalcolithic periods (9-27%, 353/3,752 burials; χ^2=11.89, p<0.001), suggesting egalitarian enforcement mechanisms preceded visible status marking. Far from being the origin of normativity, prescriptive language acted as a high-fidelity transmission technology that allowed proto-norms to scale beyond Dunbar's number (~150 individuals) and institutionalize into recognizable legal forms. The written law, therefore, is not the source of obligation but its fossilized echo. This framework validates previously documented anomalies—including the 93.3% functional extinction of the Nineteenth Amendment despite formal validity and Argentina's 93-year fiscal

constitutional lock-in—which remain inexplicable under textualist or institutionalist theories but are predicted by memetic competition dynamics. Testable predictions for cross-linguistic deontic density, ritual frequency correlations, and neural overlap studies are operationalized with specific falsification criteria.

1. Introduction: Law Without Legislators

Legal positivism assumes that norms require formulation: no rule without a rule-maker, no obligation without a command. Austin's sovereign, Kelsen's Grundnorm, Hart's rule of recognition—all presuppose that law begins with articulation. But what if the first laws were never spoken?

Consider a Pleistocene band where food sharing is enforced not by decree, but by silent ostracism; where mating taboos are maintained not through statutes, but through ritualized avoidance and bodily markers; where cooperation persists not because individuals "agree" to rules, but because deviants are excluded, punished, or ignored. These are not mere customs—they are **heteronomous norms**: behavioral regularities sustained by third-party sanctions, independent of individual consent. They impose obligations. They structure social space. They persist across generations.

This paper contends that such proto-norms constitute the **evolutionary substrate** of law. They emerged long before syntactic language, let alone writing, and functioned as adaptive solutions to the problem of cooperation in increasingly large and interdependent groups. Crucially, they did not require agents to "understand" them—only to follow and enforce them. In Dennett's terms, this is *competence without comprehension*: a system can be functional without its participants grasping its logic (Dennett, 2017). Termites build cathedral mounds without architectural blueprints. Markets coordinate millions without central planners. And Pleistocene bands enforced norms without legislators.

I further argue that these proto-norms acted as **selective pressures** that favored the evolution of prescriptive language—specifically, deontic modality (must, may, should)—which in turn enabled norms to become abstract, generalizable, and decoupled from immediate context. The result was the emergence of law as a **cultural extended phenotype**: an external structure built by normative memes to enhance their own replication and stability (Dawkins, 1982).

This reframing suggests three revisions to legal theory. First, law does not originate in sovereign will but in decentralized selection processes operating on competing behavioral patterns. Second, legal coherence is secondary to replicative fitness; "irrational" laws persist if they serve meme survival, regardless of logical elegance. Third, current legal texts cannot

be understood without reconstructing the **paleo-normative substrates** they fossilize—the deep cultural memes that predated their linguistic expression.

Critics might object that pre-linguistic norms leave no archaeological trace. But convergent lines of evidence—from primate partner selection and coalition dynamics to mortuary practices indicating egalitarian treatment, to the universal presence of deontic modality in all natural languages—suggest that proto-normativity was not merely plausible but *necessary* for the evolution of our species. No known human society lacks norms. No natural language lacks deontic structures. And no primate lineage more closely related to humans than macaques lacks proto-normative behaviors. This consilience across four independent streams (primatology, archaeology, linguistics, developmental psychology) provides strong inferential support.

This paper proceeds as follows. Section 2 presents evidence for proto-normativity from primatology (third-party punishment absence, coalition formation, partner selection), archaeology (Paleolithic egalitarian signature), and ethnography (ritual norm transmission), culminating in a fractal case study of the !Kung San. Section 3 examines the evolutionary emergence of deontic modality and its role in enabling normative scaling beyond small groups. Section 4 develops the extended phenotype framework and validates it against prior empirical findings from constitutional memory extinction and fiscal lock-in research. Section 5 details the comparative methodology underlying these claims. Section 6 addresses major objections. Section 7 explores implications for constitutional interpretation, legal transplants, and the law-morality distinction. Section 8 presents testable predictions in operationalized form. Section 9 concludes by situating law within the deep history of human cooperation[1].

2. Proto-Normativity: Norms Before Language

Evidence from primatology, paleoanthropology, and ethnography suggests that the roots of normativity lie deep in our evolutionary past—far deeper than language itself.

2.1. Primatological Evidence: The Components of Proto-Normativity

2.1.1. Absence of Third-Party Punishment in Chimpanzees

[1] Replication materials, including systematic literature reviews, quantitative data tables, and complete bibliography, are publicly available at: https://github.com/adrianlerer/Before-the-Command-Was-Spoken . All data and analyses are provided under MIT License for academic and research purposes.

Experimental studies with chimpanzees (*Pan troglodytes*), our closest living relatives, consistently show **absence of third-party punishment**—the sanctioning of norm violators by individuals who are not directly harmed. This is theoretically significant: if proto-norms operated through third-party enforcement in early hominins, we would expect to find precursors in Pan.

Ultimatum Game Evidence: In modified ultimatum game experiments where chimpanzees (N=11) could accept or reject unequal food divisions, they rejected unfair offers at rates of 5-14%, compared to approximately 50% rejection in human adults when fair alternatives are available (Jensen, Call, & Tomasello, 2007). Chimpanzees behaved as "rational maximizers," accepting any non-zero offer regardless of inequity. The contrast is stark: humans reject unfair offers even at personal cost, enforcing fairness norms; chimpanzees do not.

Controlled Punishment Paradigm: Riedl et al. (2012) tested whether chimpanzees (N=13, 60 unique triads) would punish theft when observing third parties being robbed. Using a trapdoor collapse paradigm where subjects could costly punish thieves by removing stolen food, they found dominant chimpanzees punished significantly more when *they* were robbed (second-party) than when observing others robbed (third-party): $Z = 1.995$, $p = 0.046$. Inspection revealed **no individual consistently punished third-party theft**. Kinship with the victim had no effect ($\chi^2(1) = 0.15$, $p = 0.70$), indicating that even relatedness does not trigger third-party punishment.

Policing vs. Punishment Distinction: Von Rohr et al. (2012) documented "policing" behavior in captive chimpanzees (N=11, 586 observation hours, 438 conflicts), where dominant males intervened impartially in 15.75% of conflicts. However, this reflects conflict *management* by high-ranking individuals—reducing group disruption—rather than costly *norm enforcement*. Policing serves the intervener's self-interest (maintaining stable coalitional environment), whereas third-party punishment is altruistic and directed at norm violators regardless of escalation risk. The Riedl et al. (2012) experimental findings confirm that when intervention is costly and targets non-escalated norm violations, chimpanzees do not engage.

Spite and Inequity: Jensen, Call, and Tomasello (2006) demonstrated that chimpanzees are "vengeful but not spiteful": they retaliate when personally harmed but show indifference to disadvantageous inequity when harm is not directed at them. In experimental conditions where another individual simply received more food (without stealing from the subject), chimpanzees did not collapse food tables more frequently than baseline—demonstrating absence of spite, a key component of human norm psychology.

Convergence: Four independent studies (total N≈46 chimpanzees) using different experimental paradigms (ultimatum games, theft scenarios, inequity tests, policing observations) converge on the same pattern: chimpanzees exhibit second-party punishment (retaliation when personally harmed) but not third-party punishment (sanctioning of norm violators who harmed others). This suggests third-party punishment is a **derived trait** in the human lineage, emerging after the Pan-Homo split approximately 6-7 million years ago.

2.1.2. Coalition Formation: Leveling vs. Hierarchical Patterns

If proto-norms in early human societies operated through "reverse dominance hierarchies" (Boehm, 1999)—where coalitions of subordinates suppress would-be alphas—we should find precursors in primate coalition behavior. The evidence reveals a critical distinction.

Bonobo Female Coalitions (Leveling Model): Surbeck et al. (2025) analyzed 30 years of observational data across 6 wild bonobo communities (*Pan paniscus*), documenting 1,786 intersexual conflicts. Female bonobos formed coalitions at rates varying from 1 every 11.1 days (Kokolopori-Kokoalongo) to 1 every 203.5 days (Kokolopori-Fekako), with **85% of coalitions directed against males** (not against other females). Despite being physically smaller than males (sexual dimorphism favors males), females achieved dominance: 61% of intersexual conflicts were won by females, and 70% of males were outranked by each female on average across 15 years and 5 communities.

The causal relationship between coalitions and power is robust. GLMM analysis controlling for community, year, and individual random effects showed: $\beta = 0.762$, SE = 0.146, $\mathbf{p < 0.001}$ for the effect of coalition frequency on conflicts won by females. When restricting analysis to coalitions specifically targeting males (FFCm), the effect strengthened: $\beta = 0.784$, SE = 0.152, $\mathbf{p < 0.001}$. This constitutes the strongest primate evidence for reverse dominance hierarchies: subordinates (physically smaller females) use coalitions to *invert* the dominance hierarchy that would otherwise result from individual contest competition.

Chimpanzee Male Coalitions (Hierarchical Model): In stark contrast, chimpanzee male coalitions serve to *ascend* rather than *suppress* dominance. Nishida and Hosaka (1996) documented 40 coalitions in 390 observation hours (7.1 coalitions per 100 hours) among Mahale chimpanzees (N=9 adult males). These coalitions were overwhelmingly between individuals of adjacent rank competing for higher positions. Watts (1998) found that alpha males in Ngogo maintained their position through coalitions with key allies, and losing those allies resulted in rank loss. The function is hierarchical competition, not leveling.

Mathematical Model of Leveling Coalitions: Pandit and van Schaik (2003) formalized when leveling coalitions are viable. Their N-player coalitionary game model predicts that leveling coalitions only emerge when contest potential (b) is low: $b < 0.5$. When $b \approx 1$ (despotic), dominants can monopolize resources and coalitions cannot overcome individual power asymmetries. The model correctly predicts coalition size (observed mean: 2.11-2.85 individuals) and composition (83% between adjacent-rank individuals in baboons). Critically, the model explains species variation: Papio ursinus (despotic, high b) lacks leveling coalitions, while P. anubis (more egalitarian, low b) exhibits them.

Theoretical Synthesis: The bonobo-chimpanzee contrast demonstrates that reverse dominance hierarchies are *ecologically contingent*, not universal among great apes. Bonobos' female coalitions function as proto-normative enforcement—suppressing dominance displays through coordinated sanctioning—while chimpanzee coalitions function as competitive alliances. For human evolution, this suggests that reverse dominance required specific selective pressures, likely related to obligate cooperative foraging and high dependence on shared resources where individual monopolization (high b) was ecologically constrained.

2.1.3. Social Selection as Alternative to Normative Enforcement

If chimpanzees lack third-party punishment, how do they sustain cooperation and manage free-riders? The answer lies in **social selection**: preferential partnering with cooperators and exclusion of defectors from future interactions.

Partner Choice Experiments: Melis, Hare, and Tomasello (2006a) demonstrated that chimpanzees preferentially chose collaborative partners with whom they had prior success over partners with whom collaboration had previously failed. This shows *prospective partner choice* rather than *retrospective punishment*: non-cooperators are avoided in future interactions but not actively sanctioned. In rope-pulling tasks requiring coordination, pairs with high mutual tolerance (established through independent food tolerance tests) collaborated significantly more successfully than intolerant pairs (Melis et al., 2006b).

Stag Hunt Paradigm: When does cooperation occur despite absence of punishment? Bullinger, Melis, and Tomasello (2011) constructed a Stag Hunt scenario where chimpanzees could choose between a low-reward solo option ("hare") and a high-reward cooperative option ("stag"). When a partner was present, chimpanzees bolted for the stag in approximately **90% of trials**. Cooperation emerged without communication, commitment, or punishment mechanisms. The critical factors were proximity (partner availability) and tolerance (willingness to share rewards).

Field Evidence from Meat Sharing: Wild observations confirm that chimpanzees sustain cooperation without punishing free-riders. In Taï Forest group hunts, **83% of bystanders**—individuals who did not actively participate in the chase—nonetheless obtained meat (Boesch, 1994; Melis et al., 2011). Meat access depends primarily on proximity to the kill site rather than hunting effort. When hunters attempt to monopolize meat, subordinates use harassment to obtain shares, but this is dyadic negotiation, not third-party norm enforcement. Juveniles often initiate hunts (perhaps due to impulsivity), reducing classical free-rider problems where subordinates wait for others to bear costs.

Contrast with Human Cooperation: Human cooperation scales beyond immediate partners via third-party punishment of norm violators, even when the punisher is not directly harmed and will not interact with the violator again (Fehr & Gächter, 2002). Chimpanzees, lacking this mechanism, sustain cooperation through repeated dyadic interactions with selected partners—a strategy that functions effectively in groups up to ~150 (Dunbar's number) but cannot support the large-scale, anonymous cooperation characteristic of human societies.

Theoretical Implication: Proto-normative cooperation in Pleistocene bands likely resembled chimpanzee patterns: partner selection based on reputation, tolerance among familiars, and dyadic retaliation against direct transgressors. Third-party punishment—and with it, the capacity for abstract normative rules enforced by disinterested parties—is a derived human trait that enabled cooperation to scale. The evolution of prescriptive language (Section 3) may have been driven precisely by this scaling problem: verbal articulation of rules allows *anyone* to enforce norms, not just those directly harmed or partnered.

2.1.4. Reconciliation: Phylogenetically Conserved Proto-Normative Repair

While chimpanzees lack third-party punishment, they exhibit robust reconciliation behavior—post-conflict affiliative contact between former opponents that restores social relationships. This demonstrates that proto-normative patterns of relationship management predate language.

Reconciliation occurs at substantial rates across multiple primate species: chimpanzees show Corrected Conciliatory Tendency (CCT) of 41.2% (SD=12.8%; Preuschoft et al., 2002; N=16 individuals, 298 conflicts); bonobos 22.31% (SD=23.57%; Clay & de Waal, 2013; N=36 individuals, 346 conflict pairs); and capuchin infants reach 50% (Weaver & de Waal, 2003; N=24 individuals, 268 conflicts). Reconciliation is temporally structured: de Waal and van Roosmalen (1979) found that 30% of victim contacts in the first 5 minutes post-conflict were with the former aggressor (vs. 5.3% expected by chance; $\chi^2 = 371$, $p < 0.001$).

Critically, reconciliation occurs **without linguistic formulation**. Bonobos distinguish "kiss" (mouth-to-mouth contact, characteristic of reconciliation) from "embrace" (ventro-ventral contact, characteristic of consolation) through embodied practice (de Waal & van Roosmalen, 1979). The ratio of kiss to embrace was 1.21 for reconciliation versus 0.13 for consolation ($\chi^2 = 24$, $p < 0.01$ for behavioral contingency). This demonstrates that normative behavioral differentiation—distinguishing reconciliation (repair relationship with opponent) from consolation (reduce victim's stress)—does not require symbolic representation.

Reconciliation's function is stress reduction: in both capuchins and bonobos, auto-scratching (stress indicator) decreases significantly post-reconciliation but not post-control periods (Weaver & de Waal, 2003: Wilcoxon $T = 44.0$, $p = 0.004$; Clay & de Waal, 2013: $Z = 3.99$, $p < 0.001$). Early attachment predicts reconciliation capacity: individuals raised by mothers reconcile more than orphans in both species (Clay & de Waal, 2013: $\beta = 11.196$, $SE = 2.355$, $T = 4.754$, $p < 0.001$; Weaver & de Waal, 2003: $R^2 = 0.50$, $F(2,22) = 9.88$, $p < 0.001$).

These findings demonstrate that proto-normative patterns—behavioral regularities that maintain social cohesion through patterned responses to transgression—are phylogenetically ancient, predating not only language but the Pan-Homo split. Reconciliation is a form of relationship repair that operates without articulated norms yet serves the same function: restoring cooperation after conflict.

2.2. Archaeological Evidence: The Paleolithic Egalitarian Signature

If proto-norms in Pleistocene societies operated through ostracism and exclusion—as Boehm's ethnographic evidence from hunter-gatherers suggests—we would expect to find archaeological markers of social exclusion: prone positioning in burials, absence of grave goods, spatial segregation from normative burial clusters. The archaeological record reveals the opposite.

2.2.1. Absence of Exclusionary Practices in Upper Paleolithic

Systematic review of Upper Paleolithic burial practices (35,000-10,000 BP; N≈150 individuals across 28 sites) reveals conspicuous absence of exclusionary markers. Only 1 of 150 burials (0.67%) exhibits potentially deviant characteristics, compared to 353 of 3,752 burials (9-27% depending on region) in Neolithic/Chalcolithic contexts ($\chi^2(1) = 11.89$, **$p < 0.001$** for temporal difference).

The Sunghir Case—Anti-Exclusion: Most strikingly, individuals with severe physical pathologies received the *most* elaborate burials. At Sunghir (Russia, 24,000 BP), two

children—one (Sunghir 2) with femoral bowing suggestive of skeletal dysplasia, the other (Sunghir 3) with severe chondrodysplasia limiting mobility—were interred together with over 10,000 items (Formicola & Buzhilova, 2004). Grave goods included:

- Sunghir 2: 4,903 ivory beads, 250 perforated arctic fox teeth, 2 mammoth ivory spears (2.4 m each—exceptional), 20 ivory discs, 1 ivory bracelet, 1 anthropomorphic ivory sculpture
- Sunghir 3: 5,274 ivory beads, 250 perforated fox teeth, 1 mammoth ivory spear (1.66 m), 1 perforated ivory pin

Manufacturing time is estimated at >7,500 person-hours for beads alone (approximately 4 years of full-time work), plus extensive ochre application. Comparative analysis shows Sunghir 2/3 received higher burial investment than any other documented Upper Paleolithic individuals, including the able-bodied Sunghir 1 adult (~3,000 beads). This pattern is the *opposite* of exclusion: disability did not marginalize but may have conferred special status, possibly through ritual or symbolic roles.

Dolní Věstonice Ambiguity: The triple burial at Dolní Věstonice (Czech Republic, 26,000 BP) presents the only potential Upper Paleolithic exclusionary case, yet closer analysis reveals ambiguity. The central individual (DV15) had severe congenital pathology, yet occupied the privileged central position with extensive ochre. The right individual (DV14) was buried prone—the sole prone burial in the sample—but DNA analysis (Mittnik et al., 2016) revealed all three individuals were male, contradicting earlier interpretations of sexual rivalry or childbirth death. The prone position may reflect ritual significance rather than exclusion: DV14 was oriented toward DV15, and the arrangement suggests deliberate compositional intent. Moreover, the burial included 27 fox teeth (DV14), ivory beads, and ochre—inconsistent with purely exclusionary treatment.

2.2.2. Temporal Pattern: Exclusion Emerges Post-Agriculture

The emergence of exclusionary burial practices coincides precisely with agricultural intensification and institutionalized inequality:

Neolithic Transition: Early Neolithic sites show initial weak differentiation (2.5-3.75% deviant burials in Mesolithic; Neeley, 2018), increasing to 9-15% in full Neolithic (Chapman, 2000; Rebay-Salisbury et al., 2010) and reaching 20-27% in Chalcolithic/Bronze Age tells (permanent settlements show 3× higher deviant burial frequency than dispersed sites).

Correlation with Inequality: The temporal pattern correlates with doubling of wealth Gini coefficients. Nørtoft (2022) calculated Gini for Moravian Corded Ware burials: early period G = 0.32 (comparable to Paleolithic estimates) versus late period G = 0.64. Deviant burials cluster in high-Gini contexts, suggesting exclusionary practices serve to mark and legitimize status hierarchies.

Regional Patterns: Central/Southeast European Neolithic shows highest deviant burial frequencies (20-27%), correlating with intensive cereal agriculture and tell settlements. These are precisely the contexts where archaeological evidence documents emergent social stratification, suggesting that *visible* exclusionary practices are products of scaled, stratified societies.

2.2.3. Theoretical Interpretation

Three interpretations are consistent with the data:

H1 - Expulsion Without Burial: Excluded individuals were expelled alive, and their remains never entered burial contexts. This predicts dispersed spatial signature of isolated remains rather than clustered "deviant" burials. Evidence is indirect: scattered human skeletal elements are documented at Upper Paleolithic sites (e.g., isolated crania, single long bones) but systematic analysis of their frequency and context relative to formal burials is lacking.

H2 - Face-to-Face Enforcement: Small Paleolithic bands (<50 individuals) enforced norms through direct mechanisms (partner selection, immediate retaliation, gossip) that did not require post-mortem marking. This is consistent with chimpanzee cooperation patterns (Section 2.1) and predicts that exclusionary burial practices should correlate with group size. Available data support this: tell populations (200-500 individuals) show higher deviant burial rates than dispersed settlements (<100 individuals).

H3 - Death Resets Status: Even deviants received normative burial because death "reset" social obligations, consistent with widespread ethnographic patterns where enemies are honored in death (Hertz, 1907). This predicts uniform burial treatment regardless of in-life status, with status marking emerging only when elites require permanent legitimation of hierarchy.

The data most strongly support H2 and H3: reverse dominance hierarchies enforce norms in life through immediate social mechanisms but do not require permanent markers. Only when these hierarchies *collapse* (Neolithic transition to agriculture, sedentism, and property accumulation) do elites require visible exclusion to legitimate inequality. The archaeological

signature thus reflects not the origin of normativity but the *institutionalization* of hierarchy—precisely when proto-normative egalitarian enforcement breaks down.

2.3. Ethnographic Evidence: The Ritual That Carried the Law

Among the !Kung San of the Kalahari, there is no word for "law" (Lee, 1979). Yet strict norms govern sharing, marriage, and conflict resolution. How?

Through the **"insulting the meat"** ritual: when a hunter returns with game, others deliberately belittle his catch. "You call that a kill? This animal is nothing but bones!" The successful hunter is expected to respond with humility: "Yes, I am ashamed. I don't know how I managed to hit such a scrawny thing." This performance prevents arrogance and enforces egalitarianism—not through commands, but through **embodied, repetitive practice** (Lee, 1969; Marshall, 1961).

No prescriptive language is needed. The norm—"do not boast"—is encoded in gesture, tone, and timing. Violators face not fines, but **social death**: exclusion from future cooperative hunts, withdrawal of marriage alliances, and ultimately ostracism. Wiessner (2005) documents that even minimal deviation from the ritual script triggers collective sanctioning, maintaining the pattern across generations despite its apparent "irrationality" (meat quality is objective; the insult is fiction).

This micro-system contains the entire architecture of law:

- A **behavioral rule** (anti-arrogance norm),
- A **sanction mechanism** (public shaming, exclusion),
- A **transmission device** (ritual participation by youth),
- And a **group-level function** (cooperation maintenance despite hunting variance).

It is law in its purest form: **obligation without articulation**. The written statute is merely its distant, abstract descendant. What the !Kung case reveals is that normativity does not require linguistic formulation—it requires only *patterned enforcement with social consequences*. The ritual is not decoration; it is the delivery system.

Critically, when !Kung communities experience sustained contact with agricultural or monetary economies, the "insulting the meat" ritual decays within 1-2 generations (Wiessner, 2002). This is precisely what Extended Phenotype Theory predicts: when the underlying selection pressure (egalitarianism required for mobile foraging) disappears, the phenotypic expression (ritual) loses fitness and vanishes—regardless of its formal "validity"

in oral tradition. The parallel to constitutional memory extinction is exact: the Nineteenth Amendment remained formally valid for decades after its functional extinction from judicial discourse (Section 4.2).

3. The Deontic Threshold: When Language Became Obligatory

If proto-norms worked without language, why did prescriptive language evolve?

3.1. The Scaling Problem and Dunbar's Limit

Robin Dunbar's research on primate neocortex ratio and group size reveals a consistent pattern: social group size correlates with neocortical volume across primate species, with humans' cognitive architecture supporting approximately 150 stable relationships—the "Dunbar number" (Dunbar, 1992). Beyond this threshold, gossip-based reputation tracking breaks down. You cannot remember who cheated whom in a band of 500, let alone coordinate third-party punishment based on reputation.

Yet archaeological evidence suggests post-glacial human groups exceeded 150 by the Upper Paleolithic (15,000-10,000 BP), particularly in resource-rich environments like the Nile Valley and Levantine coast. How did cooperation scale when chimpanzee-style partner selection and dyadic reputation tracking became computationally intractable?

The answer lies in **abstract, generalizable rules**. Unlike imperatives ("Grok, give me meat!"), which are agent-specific and context-bound, deontic statements ("One must share meat with kin") are **impersonal, general, and binding across situations**. They allow norms to detach from particular actors and become properties of the social world itself—what Tomasello (2009) calls "agent-neutral obligations." A verbal rule can be invoked by *anyone*, not just those directly harmed or with personal knowledge of the transgressor. This enables third-party enforcement to scale: even newcomers or those who didn't witness the violation can apply the rule.

This capacity required two cognitive prerequisites:

1. **Recursion**: The ability to embed propositions ("You must believe that sharing is required").
2. **Displacement**: Mental representation of non-present scenarios ("If you hunt tomorrow, you must share").

Both emerge with the symbolic revolution of the Upper Paleolithic (~70,000-50,000 BP), marked by representational art, long-distance trade networks, and elaborate burial practices—all requiring manipulation of abstract, displaced referents (Deacon, 1997).

3.2. Linguistic Evidence: Deontic Modality as Universal and Ontogenetically Primary

If deontic language evolved under selection pressure from proto-norms, we should observe two patterns:

Pattern 1: Linguistic Universality. Deontic modality is present in all documented natural languages, though its morphological expression varies (Palmer, 2001). English uses modal auxiliaries (*must, should, may*). Korean uses verb endings (*-eya hata* for obligation). Turkish uses particles. But the semantic category—obligation, permission, prohibition—is universal. This cross-linguistic consistency suggests deep evolutionary roots rather than recent cultural innovation.

Pattern 2: Ontogenetic Priority. If deontic modality was evolutionarily early, children should acquire it before other modal categories. Choi (2006) demonstrated that Korean children master deontic *-eya hata* ("must") significantly earlier than epistemic *-keyss* ("probably"), contradicting predictions from cognitive complexity (epistemic reasoning about beliefs should be simpler than normative reasoning about obligations). Papafragou (2006) replicated this pattern in Greek: children produced deontic modals at mean age 2;8 versus epistemic modals at 3;4 (6-month developmental gap). Ozturk and Papafragou (2007) found identical sequencing in Turkish.

This ontogenetic sequence mirrors phylogeny: obligation-marking preceded probability-marking because *proto-norms selected for it*. Groups whose members could articulate general obligations ("All hunters must share") outcompeted those relying solely on case-by-case imperatives or dyadic partner memory. The linguistic universality combined with developmental priority constitutes strong evidence that deontic structures are not arbitrary cultural inventions but evolutionary adaptations.

Metaphorical Grounding: Cross-linguistic analysis reveals that deontic modals often derive historically from spatial/force metaphors. Sweetser (1990) documents English *must* deriving from Old English *moste* ("to be compelled by force"). Similar patterns appear in unrelated languages: Mandarin *bìxū* (必须, "must") contains *xū* ("need/require"), originally a spatial containment metaphor. This suggests deontic language builds on more primitive force

schemas—consistent with proto-norms operating through physical exclusion and coercion before linguistic articulation.

3.3. The Baldwinian Amplifier: Language as Niche Construction

Prescriptive language thus functioned as a **Baldwinian amplifier**: groups with even rudimentary deontic communication could coordinate more effectively, outcompete others, and create environments that further selected for normative cognition (Baldwin, 1896; Weber & Depew, 2003). This is niche construction: the organism modifies its environment, which then selects back on the organism (Laland et al., 2000).

The feedback loop:

1. Proto-norms (partner selection, embodied ritual, dyadic retaliation) create selection pressure for better transmission technology.
2. Deontic language emerges, allowing norms to abstract and generalize.
3. Generalized norms enable larger group sizes (>150) and division of labor.
4. Larger groups create more complex normative demands (property, inheritance, trade).
5. Complex norms select for recursive syntax and metaphorical extension of deontic structures.

Language did not invent heteronomy—it **bootstrapped it to a new level of complexity**. And once deontic structures existed, they became available for other uses: commands to children, ritual formulae, political authority. But the original function was normative coordination beyond dyadic reputation tracking.

Critically, this resolves an apparent paradox: if third-party punishment is absent in chimpanzees (Section 2.1.1) yet required for large-scale human cooperation, when did it emerge? The answer: third-party punishment *co-evolved* with prescriptive language. Without linguistic articulation of rules, third-party enforcement is impossible—the enforcer cannot invoke a shared norm that exists only as distributed dyadic knowledge. With language, any individual can cite the rule and thereby legitimate their intervention. Deontic modality made third-party punishment scalable.

4. Law as Extended Phenotype: The Meme's Architecture

4.1. The Extended Phenotype Framework

Richard Dawkins' concept of the **extended phenotype** holds that genes express effects beyond the organism's body (Dawkins, 1982). A beaver's genes build not just the beaver's brain, but the dam—which then creates a new selective environment (pond) that favors dam-building genes. The dam is as much a phenotypic expression of beaver DNA as the beaver's teeth.

Dennett (2017) extends this to culture: **memes build artifacts that enhance their own survival**. A catchy tune "wants" to be hummed; a compelling ideology "wants" to be preached. These are not metaphors—they are descriptions of differential replication dynamics. Memes that construct phenotypic scaffolding (institutions, rituals, texts) replicate more successfully than memes that remain purely cognitive.

Applied to law, this yields a powerful insight:

Legal institutions are not tools humans use to control society; they are structures that normative memes build to control humans.

A taboo against incest, for example, is a meme that enhances group fitness by reducing inbreeding depression (Henrich & Henrich, 2007). It builds phenotypic extensions: kinship terminologies that make incestuous pairings linguistically unsayable, residence rules that physically separate potential partners, initiation rites that mark boundaries, and eventually penal codes with explicit prohibitions. These structures persist not because they are "just," but because they **replicate the meme that constructed them**.

The meme is the replicator. The law is its extended phenotype.

4.2. Empirical Validation from Prior Research

This framework is not speculative. It builds on prior empirical work demonstrating Extended Phenotype Theory's explanatory power across three domains where traditional legal theories fail.

Domain 1: Constitutional Memory Extinction

In prior research (Lerer, 2024), I analyzed citation patterns for the Nineteenth Amendment (women's suffrage, ratified 1920) across 15 U.S. Supreme Court cases spanning 1922-2023 involving female voting rights. Despite formal validity and continuing relevance, the Amendment suffered 93.3% functional extinction: only 1 of 15 cases cited it as primary authority. Textualism cannot explain this—the text remains unchanged. Precedent-based theories cannot explain it—the Amendment was cited regularly until the 1970s. Path

dependence cannot explain it—other amendments from the same era (Eighteenth, Twenty-First) remain active.

Extended Phenotype Theory explains it: when the underlying meme (gender equality via suffrage) loses fitness in the judicial memeplex—supplanted by Equal Protection doctrine, which applies to race and gender—its phenotypic expression (Nineteenth Amendment citations) collapses. The text remains, but the living norm is extinct. Judges unconsciously bypass the Amendment not due to error, but because the meme that would activate it no longer competes successfully in constitutional discourse.

This validates a key EPT prediction: legal texts can become functionally extinct despite formal validity when their underlying memes lose fitness. The archaeological parallel is exact: Paleolithic egalitarian practices (Section 2.2) persisted without visible markers because they were *living norms*; Neolithic exclusionary burials appeared precisely when norms required *external scaffolding* to persist against changed selection pressures.

Domain 2: Fiscal Constitutional Lock-in

Argentina maintained a national income tax for 93 years (1932-present) without constitutional authorization—globally unique persistence (Lerer, 2025). Brazil, Spain, and Chile faced similar fiscal crises and identical constitutional obstacles, yet successfully enacted formal amendments within 3-8 years. Conventional explanations—path dependence, veto players, political economy—cannot explain why *constitutional amendment* failed in Argentina when simpler legislative reform succeeded elsewhere.

Extended Phenotype Theory explains it: Argentina's income tax became a phenotypic expression of federalist identity memes embedded in the 1853 Constitution. The tax distributes revenue from wealthy provinces (Buenos Aires) to poorer ones, instantiating the federalist memeplex's core function. Attempting to formalize it via amendment threatened meme replication by making the arrangement *visible and contestable*. The memeplex triggered immune responses—judicial nullification threats, provincial mobilization, nationalist rhetoric—independent of the tax's economic efficiency.

The tax persists *because* it was never formally authorized: formalization would have destroyed the ambiguity that allowed federalist and centralist memes to coexist. This is analogous to the !Kung ritual (Section 2.3): the "insulting the meat" practice works *because* it operates below the threshold of explicit articulation. Making it a formal rule ("Article 1: All hunters shall respond 'I am ashamed' when praised") would destroy its function by eliminating the lived, embodied quality that sustains egalitarian enforcement.

Domain 3: International Law Multi-Arena Competition

Analysis of 60 transnational conflicts (2000-2025) showed systematic forum-shopping between ICJ, arbitration, and domestic courts (Lerer, 2024). Realist theories predict states ignore unfavorable rulings; liberal institutionalism predicts compliance. Neither explains the observed pattern: states *relocate disputes* to arenas where their legal memes have higher fitness, then comply with *that* ruling.

Example: In the Argentina-Uruguay Botnia pulp mill dispute (2006-2010), Argentina first litigated in ICJ (emphasizing sovereignty and treaty violation), lost, then shifted to domestic environmental litigation (emphasizing precautionary principle), achieved partial victory via injunction, then returned to arbitration (emphasizing regulatory rights). Uruguay reciprocated by forum-shopping to World Bank's IFC compliance mechanism. Both countries "complied" with favorable rulings and ignored unfavorable ones—but neither simply defected.

This pattern appeared in 44 of 60 cases (73.3%)—neither pure compliance nor pure defection, but strategic relocation to favorable selection environments. International law is not "law" in the positivist sense (no sovereign enforcer) but a **memetic fitness landscape** where competing normative claims seek advantageous institutional environments. This is precisely what EPT predicts: legal institutions are extended phenotypes of competing memeplexes, and actors navigate among institutions to find environments where their memes replicate.

Cross-Validation: These three previously documented patterns—constitutional extinction despite validity, fiscal arrangements persisting despite formal invalidity, and multi-arena forum shopping—were anomalous under standard legal theories. EPT explains all three through a single mechanism: legal institutions are extended phenotypes subject to memetic selection pressures independent of formal validity or logical coherence. The convergence of explanatory power across three independent domains constitutes strong empirical validation.

4.3. Phenotypic Expressions Across Scales

Extended Phenotype Theory reveals nested levels of phenotypic expression:

Micro-Level: Individual legal texts (statutes, constitutional provisions) are phenotypic expressions of specific memes. The Nineteenth Amendment expressed the suffrage-equality meme; when that meme was supplanted, the text became a "fossil."

Meso-Level: Institutional structures (courts, legislatures, administrative agencies) are phenotypic expressions of memeplexes about governance. Federal vs. unitary systems express different memes about power distribution; common law vs. civil law systems express different memes about precedent and codification.

Macro-Level: Entire legal traditions (Anglo-American, Continental European, Islamic, Chinese) are phenotypic expressions of civilizational memeplexes shaped by millennia of selection. Legal transplant failures (Section 7.2) occur when imported phenotypes express memes incompatible with the receiving culture's memeplex.

Evolutionary Time-Scale: The proto-norms documented in Section 2 (primate partner selection, Paleolithic egalitarian burial, !Kung ritual) are phenotypic expressions of even deeper replicators—likely genetic predispositions for cooperation, fairness sensitivity, and norm psychology that evolved over millions of years. Legal institutions inherit this substrate: they are cultural scaffolding built on biological foundations, creating a **gene-culture coevolutionary system** (Richerson & Boyd, 2005).

This multi-level structure explains why law simultaneously feels inevitable (reflecting deep biological/cultural substrates) and contingent (reflecting local memetic competition). Both are correct: law is built on universal proto-normative patterns but expressed through culturally specific institutional forms that compete for replication.

5. Comparative Method and Evidence Triangulation

Given the non-fossilization of prehistoric norms, this framework relies on **convergent validation** from four independent streams:

Stream 1: Primatology. If proto-norms exist in non-linguistic species (chimpanzees, bonobos, capuchins), parsimony suggests they preceded human language. The findings are dispositive: fairness expectations without symbolic communication (Brosnan & de Waal, 2003); reconciliation with embodied behavioral differentiation (de Waal & van Roosmalen, 1979); partner selection without articulated rules (Melis et al., 2006a); and reverse dominance in bonobos (Surbeck et al., 2025). Critically, the *absence* of third-party punishment in chimpanzees (4 independent studies, N=46) suggests this trait emerged specifically in the human lineage.

Stream 2: Developmental Psychology. If children acquire normative expectations before deontic language—as Tomasello (2009) and Schmidt & Tomasello (2012) demonstrate experimentally—this mirrors phylogeny. Three-year-olds enforce "game rules" on third

parties despite not being able to articulate why the rule exists (Schmidt & Tomasello, 2012: 63% of children intervened to prevent rule violations even when not personally affected). They have normative competence without comprehension—exactly Dennett's (2017) principle applied to early childhood.

Stream 3: Linguistic Universals. If all languages have deontic modality, acquired ontogenetically earlier than epistemic modality (Choi, 2006; Papafragou, 2006), this reveals evolutionary priority. The cross-linguistic consistency of this pattern—replicated in Korean, Greek, and Turkish using different morphological strategies—suggests deep biological roots rather than cultural convergence.

Stream 4: Archaeological Patterns. The systematic absence of exclusionary burials in Upper Paleolithic (0.67%) versus systematic presence in Neolithic/Chalcolithic (9-27%; $\chi^2=11.89$, $p<0.001$) provides material evidence of the egalitarian-to-hierarchical transition. This temporal signature cannot be explained by preservation bias (Neolithic sites are often poorer preserved than cave burials) or sample size alone (the pattern is statistically robust). The correlation with agricultural intensification and Gini coefficient doubling suggests causal mechanism: visible exclusion emerges when reverse dominance hierarchies collapse.

Consilience Logic: No single stream proves the thesis. But consilience across four methodologically independent domains—each using different data types (behavioral observation, experimental manipulation, linguistic analysis, material culture)—constitutes strong evidence. This is the logic of evolutionary reconstruction: we cannot observe Pleistocene norms directly, but we can triangulate their properties from extant traces.

Alternative Explanations Considered: Could the patterns reflect:

- *Measurement artifact*? No—multiple methodologies within each stream converge (e.g., primatology uses experiments, field observations, and mathematical models).
- *Cultural borrowing*? No—linguistic patterns appear in unrelated language families; primate patterns appear in species separated by 25+ million years of evolution.
- *Selection bias*? Partially—archaeological samples are small and geographically restricted (primarily Europe). But the temporal pattern is robust across all sampled regions, and the theoretical prediction (egalitarian proto-norms → hierarchical institutionalization) is independently motivated.

The convergence is most parsimoniously explained by a genuine evolutionary sequence: proto-norms predating language → deontic modality enabling scalable norm articulation → institutional elaboration post-agriculture.

6. Objections and Replies

Objection 1: "Norms require intentionality; animals don't have norms, just regularities."

Reply: This objection smuggles in modern human cognition as the definition of normativity. Define "norm" minimally: *regularity* + *sanction for deviation* + *third-party enforcement*. Chimpanzee coalitional behavior meets the first two criteria; bonobo female coalitions (85% targeting males, Surbeck et al., 2025) meet all three. If one insists that norms require conscious rule-following, one has defined the problem away rather than solving it. The question is precisely *how* conscious rule-following evolved—and the answer is: from proto-normative patterns that operated without it.

Moreover, the archaeological evidence (Section 2.2) shows *humans* enforcing norms without explicit articulation for tens of thousands of years. The !Kung "insulting the meat" ritual (Section 2.3) is a norm—it has sanctions (ostracism), transmission mechanisms (youth participation), and group-level functions (egalitarian enforcement)—yet it operates below the threshold of explicit propositional content. Requiring intentionality as definitional would exclude this paradigmatic example of proto-normativity.

Objection 2: "Rituals are arbitrary; they can't 'encode' norms."

Reply: Rappaport's (1979) crucial distinction: rituals are *indexes* (causally linked to what they signal), not symbols (arbitrary). The !Kung "insulting the meat" ritual is not arbitrary—it causally reduces arrogance by removing the social reward (praise) that would reinforce it. The ritual's form may vary, but its function (egalitarian enforcement) is causally connected to the behavioral outcome.

Evidence: when !Kung groups abandon mobile foraging and adopt sedentary agriculture, the ritual decays within 1-2 generations (Wiessner, 2002)—precisely because the underlying selection pressure (need to suppress dominance in resource-variable mobile foraging) disappears. If the ritual were arbitrary, its presence/absence would not correlate with subsistence strategy. The causal relationship between ritual form and normative function is direct: ritual violation threatens the mechanism maintaining the norm, which is why deviation triggers immediate sanction (Wiessner, 2005).

Objection 3: "This conflates correlation with causation—maybe language and norms co-emerged."

Reply: Likely correct—co-evolution is the expected pattern in gene-culture dynamics (Richerson & Boyd, 2005). But the *sequence* matters for testing the theory. We observe: (1) primates have proto-norms without language, (2) no species has prescriptive language without norms, (3) ontogeny recapitulates phylogeny (norm expectations before deontic language in children; Schmidt & Tomasello, 2012). This asymmetry suggests norms came first and created selection pressure for linguistic formulation, even if the subsequent co-evolution was bidirectional.

Critically, the Baldwinian mechanism (Section 3.3) predicts precisely this: initial behavioral innovations (partner selection, embodied ritual) create selection pressures for genetic changes (capacity for deontic cognition), which then enable further cultural elaboration (institutional law). This is not simple causation but feedback—consistent with all gene-culture coevolution models.

Objection 4: "Archaeological evidence is ambiguous—anomalous burials could be accidental."

Reply: Correct. Any single burial is ambiguous. But the *pattern* across multiple sites and time periods—spatial segregation + absence of grave goods + trauma consistent with punishment in Neolithic contexts, versus consistent absence of these markers in Paleolithic contexts—is unlikely to be coincidental. This is Bayesian reasoning: prior probability of punishment in hunter-gatherer societies is high (Boehm, 1999 documents reverse dominance in 48 of 48 ethnographic cases); Paleolithic archaeological patterns (egalitarian burial, care of disabled) update that prior toward proto-normative enforcement without permanent exclusion. The Neolithic shift then represents institutional collapse of reverse dominance, not origin of normativity.

Moreover, the *direction* of the anomaly is theoretically significant. If Paleolithic peoples lacked norms entirely, we would expect random variation in burial treatment. Instead, we see systematic *elevation* of disabled individuals (Sunghir) and consistent absence of exclusionary markers. This is evidence for active egalitarian enforcement, not mere absence of hierarchy.

Objection 5: "Extended Phenotype Theory is unfalsifiable—any legal phenomenon can be explained post hoc."

Reply: Section 8 provides falsifiable predictions with specified measurement protocols and falsification criteria. The theory predicts specific relationships between deontic lexicon density, ritual frequency, and norm persistence that can be tested with ethnographic and

linguistic data. It predicts neural overlap between ritual violation processing and legal judgment that can be tested with fMRI. These are prospective, falsifiable predictions, not post-hoc rationalizations.

More importantly, EPT made *successful* predictions about phenomena that were already documented but unexplained: Nineteenth Amendment extinction (predicted that constitutional provisions should collapse when underlying memes lose fitness; observed 93.3% citation collapse), Argentine fiscal lock-in (predicted that formalization threats would trigger memeplex immune responses; observed systematic amendment failure despite fiscal crisis), and international law forum-shopping (predicted multi-arena memetic competition; observed in 73.3% of cases). A theory that explains three previously anomalous patterns is not unfalsifiable—it is empirically validated.

Objection 6: "Free-rider punishment is missing in the primate data."

Reply: Correct—this is a genuine gap explicitly noted in Section 2.1.3. No primate studies document systematic punishment of individuals who fail to join coalitions (as opposed to individuals who defect during cooperation). However, the absence of evidence is not dispositive for three reasons:

First, methodologically, observational studies cannot distinguish non-participation from absence during coalition events, and experimental paradigms for coalition formation in primates are technically difficult. Second, the evidence we *do* have suggests alternative mechanisms: partner selection (Melis et al., 2006a) achieves functional equivalence to free-rider punishment by excluding non-cooperators from future benefits without costly sanctioning. Third, human hunter-gatherer societies enforce coalition participation through gossip and reputation (Boehm, 1999)—mechanisms that require language and may not leave behavioral traces in non-linguistic species.

The gap does not undermine the core thesis: proto-norms existed before language (evidenced by partner selection, reconciliation, coalition dynamics), and language enabled these patterns to scale (evidenced by deontic universals, archaeological transitions, and ethnographic ritual). Whether proto-norms included free-rider punishment specifically is an open empirical question that does not affect the broader evolutionary sequence.

7. Implications for Legal Theory

7.1. Constitutional Interpretation: The Paleo-Normative Substrate

If proto-norms shaped prescriptive language, textualist originalism faces a foundational problem. The "original public meaning" of the Eighth Amendment ("cruel and unusual punishment") cannot be determined solely from 1791 dictionaries or Founding-era statutes. It must be reconstructed from the *paleo-normative substrate*—the deep cultural memes about retribution, deterrence, and bodily integrity that predated the text.

Consider the trajectory: Colonial Americans inherited English common law norms about punishment, which themselves descended from medieval practices of public execution as both deterrent and community ritual (Foucault, 1975). These practices encoded ancient memes about status (nobles vs. commoners suffered different punishments), retribution (lex talionis), and deterrence (spectacular suffering as warning). When the Eighth Amendment prohibited "cruel and unusual punishment," it was not creating a norm ex nihilo—it was *formalizing resistance* to certain phenotypic expressions (drawing and quartering, burning at stake) while leaving others intact (hanging, whipping).

The text fossilized a moment in ongoing memetic competition between:

- **Retributive memes** (punishment as moral balancing),
- **Deterrent memes** (punishment as rational calculation),
- **Dignity memes** (emerging Enlightenment concepts of inherent human worth).

Judges interpreting the clause are not discovering timeless meaning—they are **navigating a fitness landscape** where these memes continue competing. This explains why Eighth Amendment jurisprudence is inherently unstable: textualists seek fixed meaning in a fossilized expression of fluid memetic competition. *Trop v. Dulles* (1958) invoked "evolving standards of decency" precisely because the underlying memes evolve; the text remains constant while its meaning shifts as the memetic fitness landscape changes.

Implication: Originalism is archaeology. And like archaeology, it requires reconstructing not just the artifact (text) but the entire cultural ecology that produced it. A judge asking "What did 'cruel' mean in 1791?" is asking the wrong question. The right question: "What paleo-normative substrates about suffering, status, and social order were competing for phenotypic expression in 1791, and how have those fitness landscapes shifted?"

This is not judicial activism—it is epistemic honesty about what constitutional interpretation entails. The alternative—treating 18th-century texts as containing determinate meaning independent of their memetic substrates—is a form of legal fundamentalism analogous to biblical literalism. Both ignore the evolutionary processes that generated the texts.

7.2. Legal Transplants: Meme-Environment Mismatch

The failure of Westminster parliamentary systems in post-colonial Africa is not a puzzle if we adopt EPT. The imported phenotype (written constitution establishing loyal opposition, parliamentary debate, ministerial accountability) assumed a pre-existing proto-normative substrate: adversarial debate as legitimate, loyal opposition as honorable, and transfer of power without violence as thinkable.

But many African societies evolved consensus-based leadership traditions where open opposition signaled disloyalty and consensus-seeking was enforced through extended deliberation and ritual (Wiredu, 1996). Transplanting Westminster into this meme pool created predictable dysfunction: opposition parties were treated as existential threats, not loyal partners; debate was seen as divisive, not productive; and "winner-take-all" elections violated deep egalitarian norms inherited from proto-normative enforcement patterns.

Successful adaptations—Tanzania's ujamaa socialism, Botswana's incorporation of kgotla assemblies into parliamentary procedure—worked precisely because they aligned imported phenotypes with local meme pools. Tanzania's single-party system (1961-1992) allowed consensus norms to operate within a formal parliamentary structure; Botswana's kgotla (village assembly) preserved egalitarian deliberation while adopting Westminster forms. These hybrid systems proved more stable than direct transplants (Erdmann & Simutanyi, 2003).

Generalization: Legal transplant success depends on **meme pool compatibility**, not logical coherence. A perfectly rational constitution will fail if it expresses memes that have low fitness in the receiving environment. Conversely, seemingly "irrational" legal arrangements persist because they express high-fitness memes regardless of formal invalidity.

This predicts specific outcomes:

- Transplants should succeed when source and recipient share proto-normative substrates (e.g., English common law to Anglo settler colonies).
- Transplants should fail when they conflict with deep memes about authority, reciprocity, or status (e.g., individual property rights to societies with communal land traditions).
- Hybrid institutions that blend imported forms with indigenous proto-norms should show highest stability (observed: Botswana Gini 0.53 and stable democracy; pure transplants like Uganda show higher instability).

7.3. The Law-Morality Distinction Dissolves

Hart (1961) distinguished law from morality by institutionalization: law has courts, officials, and coercive enforcement. But EPT reveals this as a difference of degree, not kind. Both are expressions of the same evolutionary process—memetic competition for behavioral control—differing only in their degree of institutionalization.

The !Kung "insulting the meat" ritual is not law because it lacks courts. But it has:

- A clear behavioral rule (anti-arrogance),
- Third-party enforcement (public shaming by non-victims),
- Sanctions (exclusion from future hunts),
- Transmission mechanisms (youth participation),
- And group-level functions (cooperation maintenance).

It is functionally identical to a statute prohibiting "disorderly conduct"—less institutionalized, but operating via the same logic: behavioral regularity sustained by sanctioned deviation. The archaeological evidence (Section 2.2) confirms that proto-normative enforcement preceded institutional law by tens of thousands of years.

This suggests that the hard boundary between law and morality is a modern invention, reflecting specific institutional developments (state monopoly on violence, written codes, professional judiciary) rather than fundamental categorical differences. Pre-state societies had norms as binding and effectively enforced as modern statutes—they simply used different phenotypic expressions (ritual rather than text, ostracism rather than incarceration).

For jurisprudence, this implies that debates over "law's special obligation" (Raz, 1979) are historically parochial. Law has no special obligation—it has historically recent *institutional* expressions of the same normative dynamics that governed cooperation for 200,000 years before writing. What makes law distinctive is not its normative force but its **scalability**: institutions enable enforcement beyond face-to-face groups. But the underlying mechanism—memes building phenotypic structures to control behavior—is identical.

8. Testable Predictions

This framework generates falsifiable hypotheses. Below I operationalize three predictions with specified measurement protocols, samples, expected results, and falsification criteria.

Prediction	Measurement	Sample	Expected Result	Falsification Criterion
P1: Deontic Density → Enforcement Complexity	Count distinct deontic morphemes per language using WALS Database (fields 71A: Semantic distinctions in modal marking, 72A: Imperative-hortative systems, 73A: Optative). Cross-reference with eHRAF ethnographic database coding for enforcement: binary (0=kin-based only, 1=third-party/institutional present).	60 pre-state societies from eHRAF with complete linguistic (WALS) + ethnographic data.	Spearman $\rho > 0.50$, $p < 0.01$. Societies with ≥3 distinct deontic markers should show third-party enforcement in ≥80% of cases. Ordinal logistic regression: OR > 2.5 for each additional deontic marker predicting institutional enforcement.	$\rho < 0.30$ OR $p > 0.05$ OR no difference in enforcement rates between high-deontic (≥3 markers) and low-deontic (≤1 marker) groups (χ^2 test $p > 0.10$).
P2: Ritual Frequency → Norm Persistence	Ethnographic record: count annual rituals with explicit normative content (prohibitions, obligations, sanctions) from detailed ethnographies. Norm persistence: coded as *generations without major change* (1 gen = 25 years), requiring ≥2 ethnographic observations separated by ≥25 years documenting same core norms. Use only societies with ≥50 years continuous documentation.	30 hunter-gatherer societies in eHRAF with longitudinal data spanning ≥50 years (e.g., !Kung, Hadza, Ache, Mbuti)	High ritual frequency (≥10 normative rituals/year) → norm persistence across 3+ generations in ≥75% of cases. Low ritual frequency (<5/year) → persistence in ≤40% of cases. Fisher's exact test or χ^2 test, $p < 0.05$. Linear regression: each additional ritual/year increases persistence probability by ≥5 percentage points.	No significant difference between high/low ritual groups ($p > 0.10$) OR inverse relationship (high ritual → *lower* persistence) OR effect size <2 percentage points per ritual/year.
P3: Neural Overlap (Ritual/Law Processing)	fMRI: measure BOLD activation during (a) viewing video of ritual norm violation (culturally familiar to participants, e.g., Eucharist desecration	40 participants: 20 with strong religious background (≥1 ritual participation/month for ≥10 years), 20 secular (≤4 ritual participations/year).	Mean overlap ≥60% in ACC + TPJ across both groups, reflecting domain-general norm processing. No significant difference between	Overlap <40% in either ROI OR religious/secular groups differ significantly ($p < 0.05$, Cohen's $d > 0.6$), suggesting

Prediction	Measurement	Sample	Expected Result	Falsification Criterion
	for Catholics), (b) judging legal case violation (criminal statute, e.g., theft scenario). Regions of interest: anterior cingulate cortex (conflict detection), temporoparietal junction (norm violation detection), ventromedial prefrontal cortex (social value). Overlap = percentage of voxels activated >2 SD above baseline in both conditions within each ROI.	Groups matched for age, education, and legal knowledge (law students excluded). Pre-registered analysis plan.	religious/secular groups (independent samples t-test, $p > 0.10$), suggesting shared neural substrate rather than religion-specific circuitry. Within-subjects correlation: individual overlap scores should correlate $r > 0.50$ between ritual and legal conditions.	separate cognitive systems. OR within-subjects correlation $r < 0.30$, indicating domain-specific rather than domain-general processing.

Additional Predictions (for future research)

P4: Cross-Linguistic Metaphor Patterns: If deontic modality evolved from proto-norms about physical force and spatial constraint, languages should systematically map spatial/force metaphors onto obligation. Lakoff & Johnson (1980) document this in English ("must" from "compelled by force"); systematic cross-linguistic survey across 100+ languages from 20+ language families should reveal >70% show spatial/force → deontic metaphor pathways, with no alternative metaphorical source more frequent. Falsification: <50% show this pattern, or alternative source (e.g., desire metaphors) equally or more common.

P5: Computational Agent-Based Modeling: Simulate norm evolution in populations of 10-500 agents under varying conditions: (a) proto-norms only (partner selection + dyadic punishment), (b) proto-norms + prescriptive language (generalized rules), (c) proto-norms + language + institutions (third-party enforcement by specialists). Prediction: cooperation should scale to ~150 agents in condition (a), ~500 in condition (b), and >1000 in condition (c). Mean cooperation rate should show threshold effects at these group sizes. Falsification: no group size thresholds, or language/institutions provide no scaling advantage beyond proto-norms alone.

P6: Longitudinal Ritual Decay: If rituals are phenotypic expressions of memes under selection, ritual decay rates should correlate with environmental change rates. Hunter-gatherer groups transitioning to agriculture should lose normative rituals at rate >50% within 2 generations (predicted by !Kung data; Wiessner, 2002). Groups maintaining subsistence strategy should retain >80% of rituals across 3+ generations. Falsification: no correlation between subsistence change and ritual loss, or agriculture increases ritual frequency.

9. Conclusion: The Law That Built Us

We have long believed that humans invented law to tame chaos. The evolutionary record suggests the opposite: **law invented us**—or rather, the memes that became law shaped the cognitive and social niches in which *Homo sapiens* thrived.

The first laws were not inscribed on stone tablets, but etched into behavior, ritual, and exclusion. Cooperation norms predated the capacity to articulate them. Sanctions predated statutes. Obligation predated language. Chimpanzees select partners based on prior cooperation (90% Stag Hunt success when partners present; Bullinger et al., 2011), yet lack third-party punishment (0% in costly scenarios; Riedl et al., 2012). Bonobos form female coalitions that invert dominance hierarchies (85% targeting males, $\beta=0.762$, $p<0.001$; Surbeck et al., 2025), demonstrating that reverse dominance is evolutionarily plausible though ecologically contingent. Upper Paleolithic burials show systematic absence of exclusionary practices (0.67% deviant burials), with disabled individuals receiving the *most* elaborate treatment (>7,500 person-hours at Sunghir; Formicola & Buzhilova, 2004), suggesting active egalitarian enforcement without permanent status marking.

And when language evolved, it did not create these norms—it gave them *voice*. All natural languages possess deontic modality (Palmer, 2001), acquired ontogenetically before epistemic modality (Choi, 2006; Papafragou, 2006), suggesting deep evolutionary roots. Prescriptive language was not the origin of normativity but its amplification technology. It allowed proto-norms that operated through embodied practice and immediate sanction to abstract, generalize, and scale beyond Dunbar's number. The written law is not the source of obligation but its **fossilized echo**—a phenotypic expression of behavioral regularities that persisted for hundreds of thousands of years before anyone could write them down.

This reframing has practical consequences. For legal practitioners, it suggests a diagnostic question: when a statute fails despite enforcement resources, ask not "Was it badly drafted?" but "What proto-normative meme is it competing against?" Constitutional provisions that conflict with deep meme pools (egalitarianism, kinship loyalty, reciprocity expectations) will fail regardless of textual clarity. The Nineteenth Amendment's 93.3% functional extinction

(despite formal validity) occurred because the underlying meme was supplanted by Equal Protection doctrine—a fitness competition, not a drafting error. Conversely, provisions that align with pre-existing memes persist even when formally violated: Argentina's unauthorized income tax survives 93 years because it expresses federalist identity memes embedded in the constitutional substrate.

For constitutional designers, durability depends less on logical coherence than on **meme pool compatibility**—alignment with the deep behavioral regularities that no text can erase. The most elegant constitution will collapse if it expresses low-fitness memes (Westminster parliamentary systems in consensus-based African societies). The most incoherent arrangement will persist if it expresses high-fitness ones (Switzerland's cantonal heterogeneity, persisting 700+ years despite apparent inefficiency).

And for judges interpreting vague constitutional clauses, the search for "original meaning" is archaeological work. The Eighth Amendment's prohibition on "cruel and unusual punishment" cannot be understood without reconstructing the paleo-normative substrates—memes about retribution, status, and bodily integrity—that competed for expression in 1791 and continue competing today. Textualism treats 18th-century English as containing determinate meaning; EPT recognizes it as fossilizing a moment in ongoing memetic evolution. The difference matters: one approach seeks fixed answers in static text, the other navigates fitness landscapes where meaning necessarily shifts as selective pressures change.

To study law, then, is not merely to interpret texts. It is to **excavate the deep history of human cooperation**—one fossilized norm at a time. We discover artifacts (constitutional provisions, statutes, precedents) and attempt to reconstruct the behavioral patterns they once served. Sometimes the pattern survives, and the law remains living (property rights, contract enforcement, kinship prohibitions). Sometimes the pattern dies, and the law becomes fossil (Nineteenth Amendment, blasphemy statutes, sumptuary laws). And sometimes new patterns emerge, demanding new phenotypic expressions (data privacy, algorithmic accountability, climate obligations).

In that excavation, we discover that law is not something we created to govern ourselves. It is something that created us. The proto-norms that would become law shaped the selection pressures under which *Homo sapiens* evolved: favoring individuals capable of tracking reputation, enforcing third-party sanctions, and—eventually—articulating general rules that allowed cooperation to scale beyond immediate kin. We are the descendants of those who could navigate normative fitness landscapes. And the landscapes we navigate today—constitutional democracies, international legal regimes, regulatory states—are the extended

phenotypes of memes that have been replicating, mutating, and competing for 200,000 years.

The law is older than language. It is older than Homo sapiens. It runs in our bones before it runs in our courts. And understanding it requires not legal hermeneutics alone, but evolutionary biology, primatology, archaeology, and linguistics—the full consilience of sciences that illuminate how cooperation emerged, stabilized, and scaled from small foraging bands to global civilization. This paper has attempted that consilience. The hypothesis stands subject to empirical testing: the predictions in Section 8 are operationalized, the data are replicable, and the theory is falsifiable. If it survives scrutiny, it will be because the evidence—from four independent disciplinary streams—converges on a single evolutionary trajectory. If it fails, that failure will itself be informative about which aspects of proto-normativity we have misunderstood.

Law is not written to constrain us. We are written to constrain it.

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